A
Project Report
On

‘LIFEFLOW BLOOD DONATION MANAGEMENT SYSTEM

Submitted
to

Gokul Global University, Sidhpur

For the Degree of

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY (SEM-VI)

Submitted By

GOVIND DESAI
(Enrollment No.:23001102022)
DHRUV SOLANKI
(Enrollment No.:23001102031)

Guided By

Mr. Rajan Sir
Assistant Professor
Faculty Of Computer Science & Applications
Gokul Global University, Sidhpur

Gokul College of Computer Science & Applications
Gokul Golbal University
Opp. I.O.C Depot, Siddhpur-384151

MAY-2026

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CERTIFICATE

This is to certify that **Dhruv Solanki (Enrollment No. 23001102031) And Govind Desai (Enrollment No. 23001102022)** of **Bachelor Of Science In Information Technology Semester** 6th in the **Faculty of Computer Science & Applications** during the academic year 2025-26 at **The GOKUL GLOBAL UNIVERSITY SIDHPUR**.

They have carried out the project Entitled: - "**LIFEFLOW BLOOD DONATION MANAGEMENT SYSTEM** " They are sincere and regular student as per my knowledge.

Date:

Place:

Internal Guide

Dean of

Faculty of Computer Science & Applications

<u>Declaration</u>

We hereby **<u>Govind Desai & Dhruv Solanki</u>** declare that the project entitled ''*LIFEFLOW BLOOD DONATION MANAGEMENT SYSTEM*'' is an original work carried out by us under the guidance of **<u>Mr. Rajan Sir</u>** for the part fulfilment of the award of the Bachelor Of Science In Information Technology in Faculty of Computer Science And Applications, Gokul Global University, Sidhpur.

The Project work has been carried out from 01/01/2026 to 30/04/2026

The matter embodied in this Project has not been submitted anywhere else for the award of any other Degree/Diploma.

Date:

Place:

Signature

GOVIND DESAI

DHRUV SOLANKI

ACKNOWLEDGEMENT

We wish to express our deep sense of gratitude to our guide **Mr. Rajan Sir** for introducing the Present topic as project for us and for his inspiring guidance and valuable suggestion throughout this project work. His efforts have provided us an insight of multifarious. His advice and attitude as a guide is commendable.

We are thankful to our Sir **Mr. Rajan Sir**, Faculty Of Computer Science & Applications for giving us the opportunity to work in the department. We would like to express our thanks to all the member of Faculty of Computer Science & Applications for any kind of help and support during the project work. Thanks all our dear friends for their In last but not the least we would like continuous help and support.

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1. Project Profile — LifeFlow Blood Donation Platform

LifeFlow is a full-stack, AI-powered blood donation management web platform designed to bridge the critical gap between blood donors and recipients. The platform connects donors, recipients, hospitals, blood banks, and organizations in real time, addressing the persistent crisis of blood shortages in the healthcare ecosystem. Built using React 19, Node.js, MySQL, and JWT-based authentication, LifeFlow is a final-year BSc IT project submission for the academic year 2025–2026.

Project Name	LifeFlow Blood Donation Management System
Project Type	Full-Stack Web Application (React 19 + Node.js + MySQL)
Developers	Govind Desai (Enrollment: 23001102022) Dhruv Solanki (Enrollment: 23001102031)
Academic Year	2025–2026 | BSc IT Semester VI
University	Gokul Global University, Sidhpur
Technologies	React 19 · Node.js · Express v5 · MySQL · Sequelize ORM · Prisma · JWT · Leaflet.js
Key Highlights	50+ Verified Hospitals · 8 Blood Types · 19 Features · 3 User Roles

2. Functional Requirement Specification

The following functional requirements define what the LifeFlow system must accomplish to fulfil the needs of all stakeholders — donors, recipients, hospital staff, and administrators.

ID	Feature	Description
FR-01	User Registration & Login	Donors, recipients and hospital staff can register and log in securely using JWT-based authentic
FR-02	Blood Donation Scheduling	Donors can browse active donation camps and book time slots via an interactive map.
FR-03	Real-Time Inventory	Hospitals can view and update live blood-group inventory including urgent request alerts.
FR-04	Emergency Request Alerts	Recipients can submit emergency blood requests; matching donors receive push notifications.
FR-05	AI Chatbot	An embedded AI assistant answers donor queries, eligibility questions, and camp information.
FR-06	Blood Compatibility Checker	Users can check cross-compatibility between any two blood types before requesting a transfusio
FR-07	Gamified Badge System	Donors earn XP points and unlock tier badges (Starter → Legend) after each donation.
FR-08	Community Hub	Donors share stories, read awareness articles, and interact via likes, comments, and shares.
FR-09	Admin Dashboard	Administrators manage users, hospitals, camps, inventory, and reports from a unified panel.
FR-10	Leaderboard	A public leaderboard ranks top donors by XP, fostering community motivation.

3. Module Specification

• Authentication Module

Manages user registration, login, JWT token generation with Bcrypt password hashing, and role-based access control for donors, recipients, hospital admins, and super-admins.

• Home & Discovery Module

Provides the landing page with a live ticker (real-time alerts), featured hospitals, emergency indicators, and navigation to all sub-modules. Animated with Framer Motion.

• Donation Camps Module

Displays interactive React-Leaflet maps with all active camps, real-time slot counters, search filters by city/hospital, and one-click booking via Axios API calls.

• Heroes / Gamification Module

Tracks XP points per donation, assigns tier badges (7 levels), and maintains a public leaderboard updated in real time — state managed via Zustand.

• Donor Dashboard Module

Shows each donor's donation history, earned badges, XP rank, percentile standing, and a personal impact counter (lives saved). Data fetched from MySQL via Sequelize.

• Blood Analytics Module

Displays live blood-group inventory per hospital, active donor counts, regional critical needs, and supply–demand trend charts pulled from MySQL tables.

• Emergency Module

Allows recipients to post urgent blood requests; the system queries MySQL for compatible donors and triggers instant in-app notifications.

• Community Hub Module

A social feed for donors to share personal stories with blood-group tags; supports likes, comments, and awareness news articles. Built with React 19 + Framer Motion.

• Admin Panel Module

A full management interface for super-admins to manage users, hospital registrations, camp approvals, inventory edits, and generate reports from MySQL data.

4. User Specification

LifeFlow supports four distinct user roles, each with specific privileges and workflows:

Role	Description	Key Permissions
Donor	Registered individuals who donate blood voluntarily	Register · Book camp slots · Earn XP · View badges · Post stories
Recipient	Patients or caregivers seeking compatible blood units	Search inventory · Submit emergency requests · View camp locations
Hospital Admin	Staff managing a registered hospital's data	Update inventory · Approve camps · View requests · Manage registration
Super Admin	Platform administrator with full system access.	Manage all users · Approve hospitals · View analytics · Generate reports

5. About the System

LifeFlow is a cloud-ready full-stack web application that serves as a centralised blood donation ecosystem. Built on React 19 (frontend), Node.js + Express v5 (backend), and MySQL with Sequelize ORM and Prisma Schema (database), LifeFlow integrates real-time data streams, geolocation services, and community engagement into a single cohesive platform. The system operates across desktop and mobile browsers without requiring a native app installation.

- Real-time blood inventory tracking with hospital-level granularity via MySQL + Sequelize
- Interactive React-Leaflet donation camp maps with live slot availability
- 7-tier gamified badge system to reward and retain active donors
- Smooth UI animations and transitions powered by Framer Motion
- Blood compatibility matrix covering all 8 major blood groups
- Community hub with awareness articles and peer story sharing
- Emergency alert system with donor-recipient smart matching
- Zustand-powered global state for seamless cross-component data sharing
- Vite-optimised build for fast load times and hot module replacement
- Comprehensive analytics dashboard for administrators

6. About the Existing System

Before LifeFlow, blood donation management in India relied predominantly on manual processes, phone calls, and fragmented hospital databases. The following limitations characterised the existing ecosystem:

- No centralized platform connecting donors, hospitals, and recipients simultaneously
- Blood shortage data updated manually with significant delays
- No real-time emergency matching — recipients depended on personal networks
- Lack of donor motivation mechanisms — no rewards, tracking, or engagement tools
- Camp schedules shared through paper notices or informal social media posts
- No AI assistance for donor eligibility or compatibility guidance
- Data silos between hospitals leading to duplication and inefficiency

7. Need for the New System

The healthcare sector in India faces a perennial blood shortage crisis. According to WHO reports, the country requires approximately 15 million units of blood annually, of which a significant deficit remains. LifeFlow addresses this through:

- Centralisation: A single platform for donors, hospitals, and recipients eliminates fragmentation
- Speed: Real-time inventory and emergency alerts reduce critical response times from hours to minutes
- Retention: Gamification (XP, badges, leaderboards) increases repeat donor participation
- Reach: Interactive maps and AI chatbot reduce geographic and information barriers
- Safety: Blood compatibility checker prevents mismatched transfusions
- Awareness: Community hub and news panel drive public education on blood donation

8. Technical Requirement Specification

Category	Technology / Specification
Frontend Framework	React 19 (with Hooks, React Router v7, Framer Motion animations)
Styling	Tailwind CSS — utility-first CSS framework
Mapping Library	Leaflet.js + React-Leaflet for interactive donation camp maps
State Management	Zustand — lightweight global state management
HTTP Client	Axios — promise-based HTTP client for API calls
Build Tool	Vite — fast frontend build tool and dev server
Backend Runtime	Node.js with Express.js v5 REST API
Authentication	JSON Web Tokens (JWT) + Bcrypt password hashing
Database	MySQL — relational database management system
ORM / Schema	Sequelize ORM + Prisma Schema + mysql2 Driver
API Architecture	RESTful APIs with JSON payloads; JWT Bearer token authentication
Version Control	Git with GitHub repository
Deployment	Vercel (Frontend) · Railway / Render (Backend) · PlanetScale (MySQL)

9. Hardware Requirement

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 / Ryzen 3	Intel Core i5/i7 or Ryzen 5/7
RAM	4 GB	8 GB or more
Storage	20 GB free disk space	SSD with 50 GB free
Network	Broadband (10 Mbps)	Stable 50 Mbps+ connection
Display	1280 × 720 resolution	1920 × 1080 Full HD
Server (Deploy)	1 vCPU, 512 MB RAM VPS	2 vCPU, 2 GB RAM cloud instance

10. Software Requirement

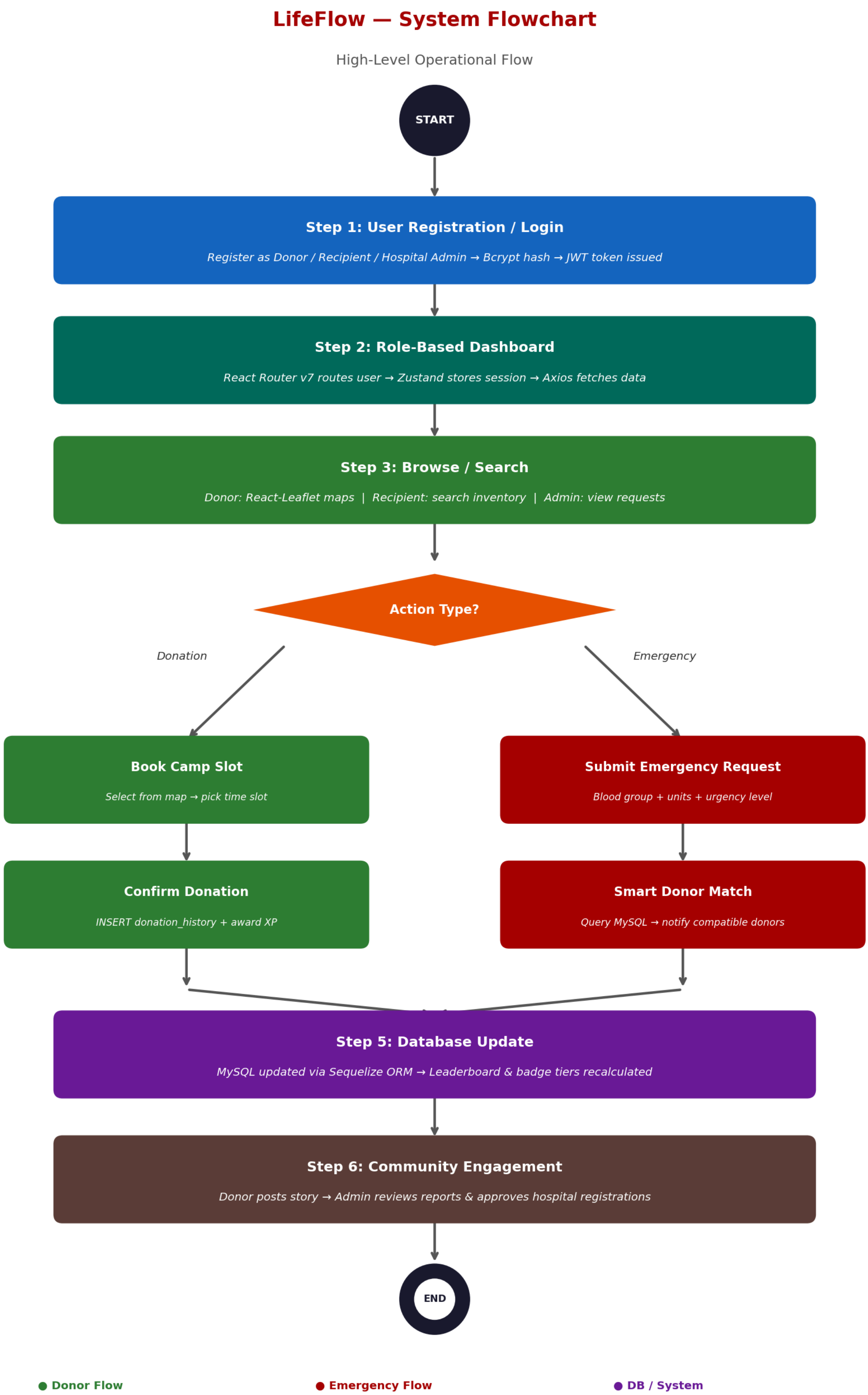
Software / Tool	Version / Details	Purpose
Node.js	v20 LTS or higher	Backend runtime environment
React	v19.x	Frontend UI library
Express.js	v5.x	Backend web framework
MySQL	v8.x	Relational database management system
Sequelize ORM	v6.x	MySQL object-relational mapping
Prisma	Latest	Schema definition & DB migrations
mysql2	Latest	MySQL driver for Node.js
JWT (jsonwebtoken)	v9.x	Stateless authentication tokens
Bcrypt	v5.x	Password hashing & verification
Vite	v5.x	Frontend build tool & dev server
Tailwind CSS	v3.x	Utility-first CSS styling
Framer Motion	Latest	Animations & UI transitions
Leaflet / React-Leaflet	Latest	Interactive donation camp maps
Zustand	Latest	Global state management
Axios	Latest	HTTP client for API requests
React Router	v7.x	Client-side routing
Git / GitHub	Latest	Source code version control
VS Code	Latest	Development IDE
Postman	Latest	API testing & documentation
Web Browser	Chrome / Firefox (latest)	Application runtime for users

11. System Flow Chart

The following describes the high-level operational flow of the LifeFlow system from user entry to the completion of a donation or emergency response cycle:

Step 1	**User Registration / Login** A new user registers as Donor, Recipient, or Hospital Admin. Password is hashed with Bcrypt, credentials stored in MySQL, and a JWT token is issued via Express v5 API upon successful login.
Step 2	**Role-Based Dashboard** React Router v7 routes the authenticated user to their personalised dashboard. Zustand stores the session state; Axios fetches dashboard data from the Node.js REST API.
Step 3	**Browse / Search** Donors browse React-Leaflet camp maps; Recipients search MySQL blood inventory by group and location; Hospitals view incoming requests from their admin panel.
Step 4	**Action Execution** Donor books a camp slot → confirms donation → XP awarded and written to MySQL via Sequelize. Recipient submits emergency request → system queries MySQL for compatible donors → notifications sent.
Step 5	**Database Update** MySQL tables are updated instantly via Sequelize ORM. Prisma schema migrations keep the DB structure in sync. Leaderboard and badge tiers are recalculated and reflected on next fetch.
Step 6	**Community Engagement** Donor optionally posts a story to the Community Hub using the React 19 form (animated with Framer Motion). Admin reviews reports and approves new hospital registrations from the admin panel.

Section 11: System Flowchart ? LifeFlow Operational Flow



12. UML Diagrams

The system design is captured through the following UML diagrams. Descriptions are provided for each diagram type used in LifeFlow.

12.1 Use Case Diagram

Illustrates interactions between actors (Donor, Recipient, Hospital Admin, Super Admin) and system use cases such as Register, Donate, Request Blood, Manage Inventory, and View Reports.

12.2 Class Diagram

Depicts the object-oriented structure of LifeFlow — key classes include User, Donor, Recipient, Hospital, BloodBank, DonationCamp, EmergencyRequest, Badge, and CommunityPost with their attributes and relationships.

12.3 Sequence Diagram

Shows the time-ordered message flow for core operations: (a) Donor Login & Camp Booking, (b) Emergency Blood Request & Notification, (c) Admin Approving a Hospital Registration.

12.4 Activity Diagram

Describes step-by-step workflows for Donor Registration, Emergency Request Resolution, and Blood Inventory Update processes.

12.5 ER Diagram

Maps the MySQL table relationships: users ↔ donation_history, hospitals ↔ blood_inventory, emergency_requests ↔ matched_donors, community_posts ↔ comments — all enforced via foreign key constraints.

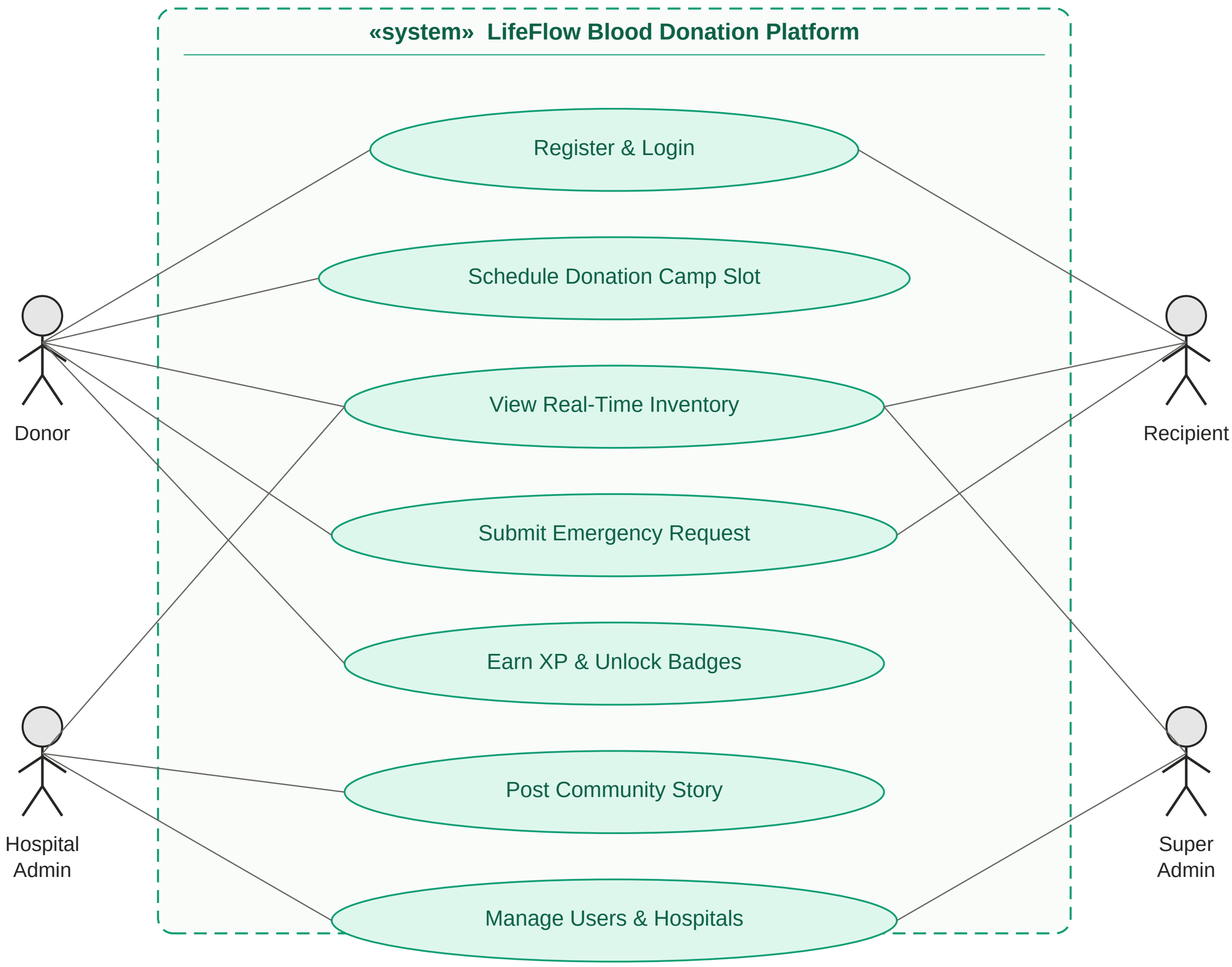
12.6 Component Diagram

Shows the architectural breakdown into React 19 Frontend (Vite), Express v5 API Server, MySQL Database (Sequelize + Prisma), and the Leaflet.js map component.

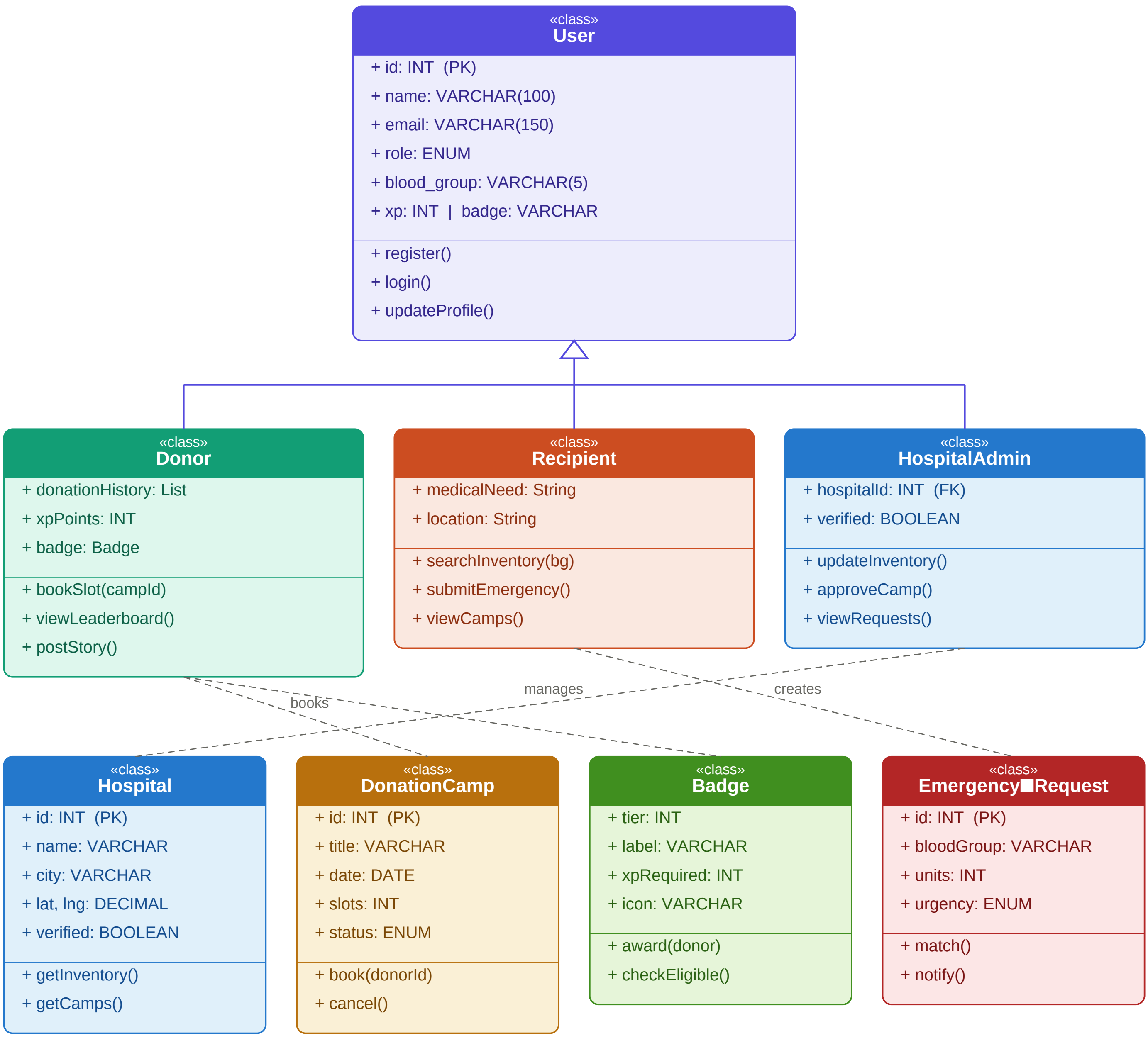
12.7 Deployment Diagram

Captures the physical deployment: Client Browser → Vercel CDN (React 19) → Railway/Render (Node.js + Express v5 API) → PlanetScale / Railway MySQL (Database).

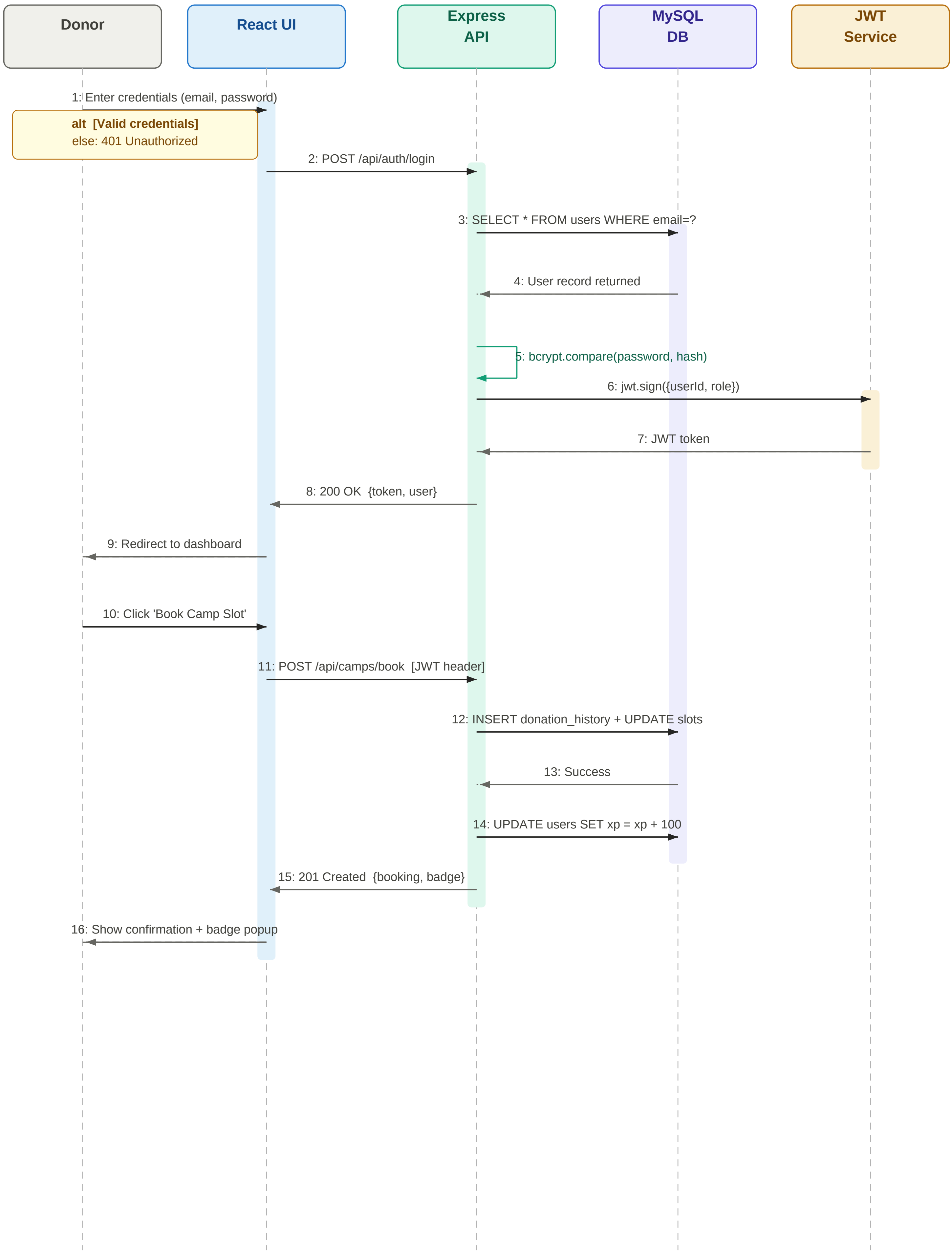
12.1 Use Case Diagram



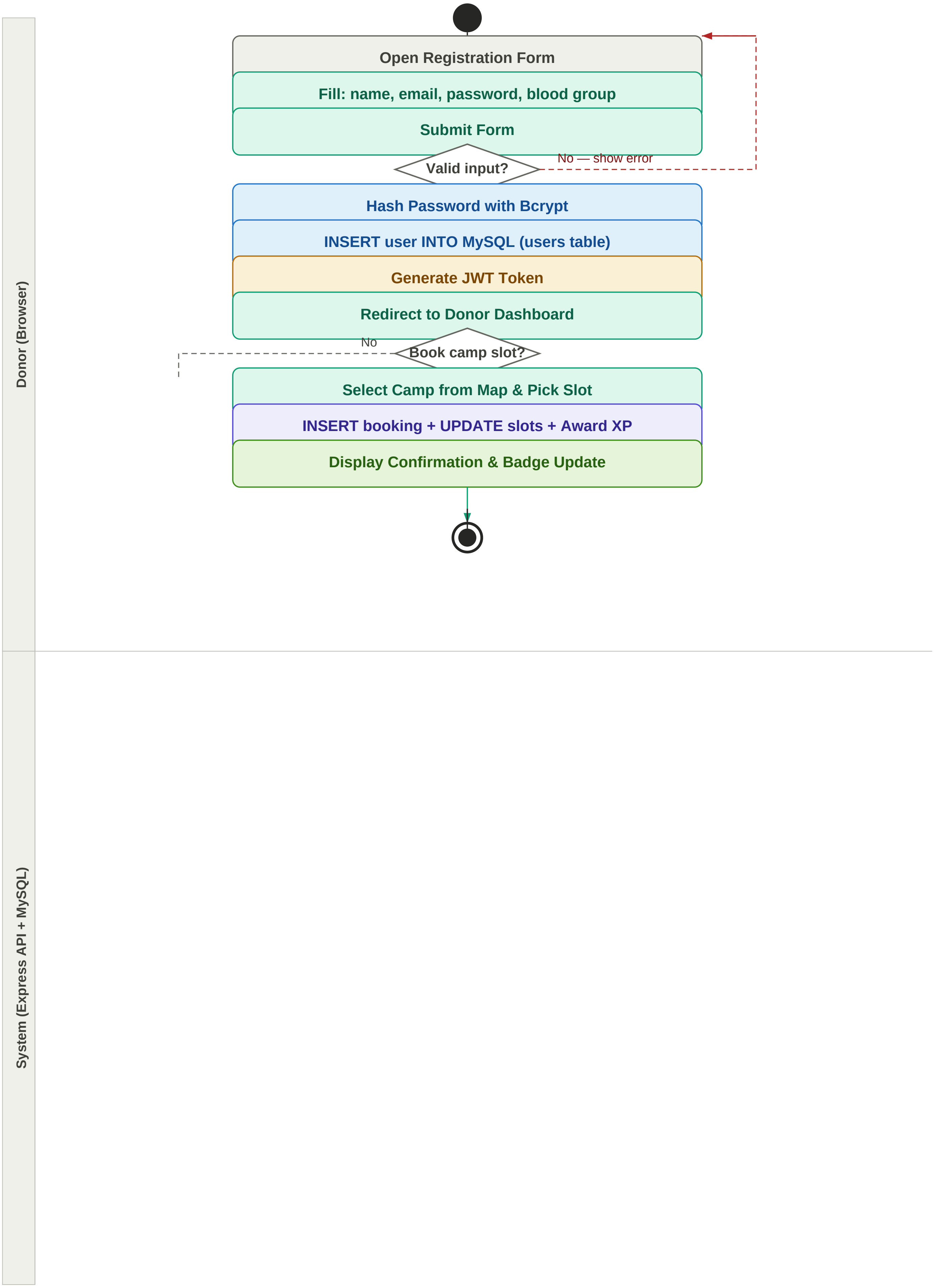
12.2 Class Diagram



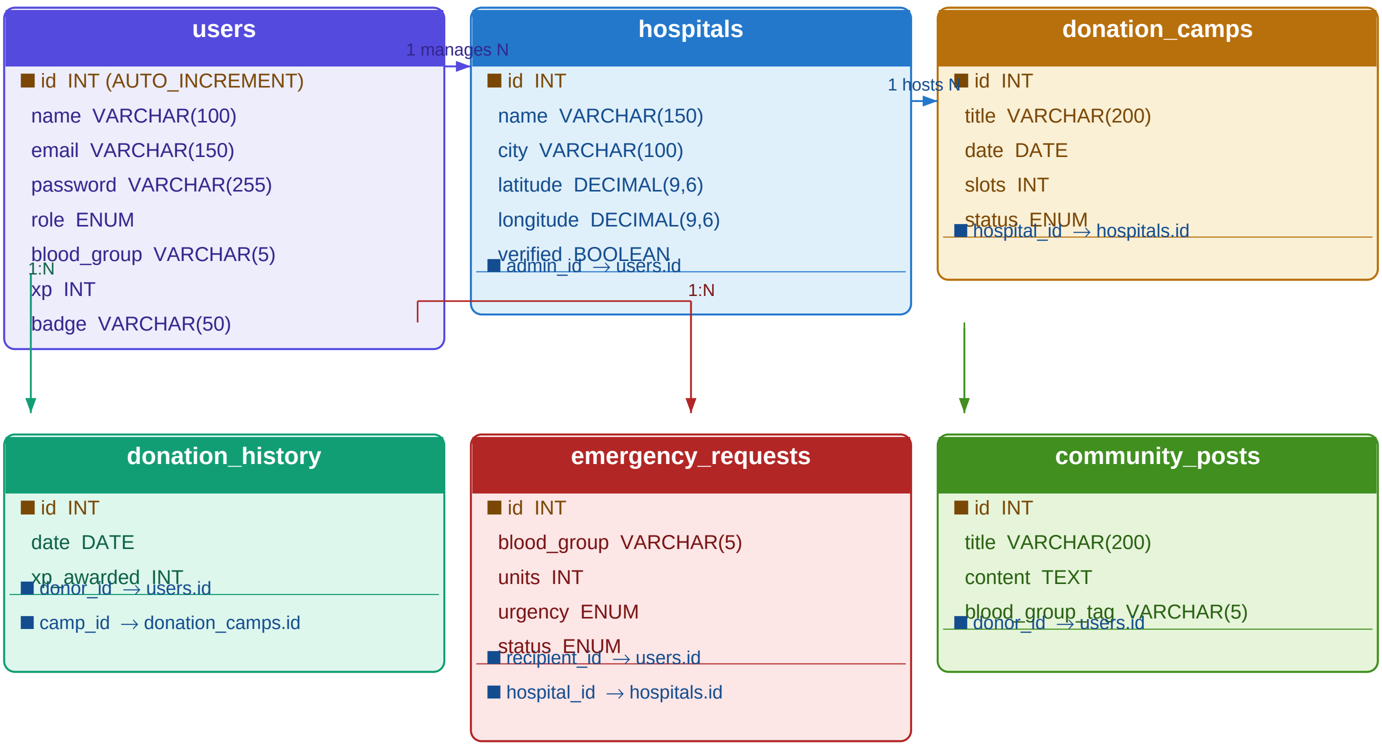
12.3 Sequence Diagram — Donor Login & Camp Booking



12.4 Activity Diagram — Donor Registration & Donation Flow



12.5 ER Diagram — MySQL Table Relationships



Legend: ■ Primary Key (PK) ■ Foreign Key (FK → references) ■■■ 1:N Relationship ENUM = predefined value set

Section 12: UML Diagrams — 12.6 & 12.7 Component & Deployment Diagrams

12.6 Component Diagram — System Architecture

React 19 Frontend	Express v5 API Server	Sequelize ORM + Prisma	MySQL Database
• Vite 5 build tool	• Node.js v20	• Sequelize v6	• users table
• Tailwind CSS	• JWT auth	• Prisma schema	• hospitals table
• React Router v7	• Bcrypt hashing	• mysql2 driver	• donation_camps
• Framer Motion	• RESTful endpoints	• Migrations	• emergency_requests
• Leaflet.js maps	• CORS + Helmet	• Model definitions	• community_posts
			• donation_history

HTTP/Axios calls → Model queries → SQL reads/writes →

External Services (planned): Email (future) · Twilio SMS alerts (future) · Map tiles (Leaflet CDN)

12.7 Deployment Diagram — Production Infrastructure

Client Device (Browser / Mobile)	Vercel CDN (Frontend Host)	Railway / Render (Backend Server)	PlanetScale (MySQL Cloud DB)
• React 19 SPA	• Static file serving	• Node.js v20 LTS	• MySQL 8.x managed
• Chrome/Firefox/Safari	• Global edge network	• Express v5 API	• Sequelize + Prisma
• Mobile responsive	• HTTPS + TLS 1.3	• JWT auth layer	• Auto-scaling

HTTPS → → REST API → SQL/ORM →

External APIs (Future / Current)

Twilio SMS alerts · NBTC Gov database · Leaflet map tiles

13. Data Dictionary

Key MySQL tables and their fields are defined below (Sequelize / Prisma schema):

Table: users

Column	Type	Description
id	INT (PK, AI)	Auto-increment primary key
name	VARCHAR(100)	Full name of the user
email	VARCHAR(150)	Unique email address (login credential)
password	VARCHAR(255)	Bcrypt-hashed password
role	ENUM	donor | recipient | hospital_admin | super_admin
blood_group	VARCHAR(5)	Blood group (A+, A-, B+, B-, O+, O-, AB+, AB-)
xp	INT	Total XP points earned through donations
badge	VARCHAR(50)	Current badge tier (Starter → Legend)
created_at	TIMESTAMP	Account creation timestamp

Table: hospitals

Column	Type	Description
id	INT (PK, AI)	Unique hospital identifier
name	VARCHAR(150)	Official hospital name
city	VARCHAR(100)	City where the hospital is located
latitude	DECIMAL(9,6)	GPS latitude coordinate
longitude	DECIMAL(9,6)	GPS longitude coordinate
verified	BOOLEAN	Admin-verified status flag
admin_id	INT (FK)	Foreign key → users.id (hospital admin)

Table: donation_camps

Column	Type	Description
id	INT (PK, AI)	Unique camp identifier
title	VARCHAR(200)	Camp name / event title
hospital_id	INT (FK)	Foreign key → hospitals.id
date	DATE	Scheduled camp date
slots	INT	Total available donor slots
status	ENUM	active | completed | cancelled

Section 13: Data Dictionary — MySQL Table Relationships

13. Data Dictionary — Key MySQL Tables

users
■ id INT (PK, AI)
name VARCHAR(100)
email VARCHAR(150)
password VARCHAR(255)
role ENUM
blood_group VARCHAR(5)
xp INT
badge VARCHAR(50)
created_at TIMESTAMP

hospitals
■ id INT (PK, AI)
name VARCHAR(150)
city VARCHAR(100)
latitude DECIMAL(9,6)
longitude DECIMAL(9,6)
verified BOOLEAN
■■ admin_id → users.id

donation_camps
■ id INT (PK, AI)
title VARCHAR(200)
■■ hospital_id → hospitals.id
date DATE
slots INT
status ENUM
(active|completed|cancelled)

↓ 1:N   ↓ 1:N   ↓ 1:N

donation_history
■ id INT (PK, AI)
date DATE
xp_awarded INT
■■ donor_id → users.id
■■ camp_id → donation_camps.id

emergency_requests
■ id INT (PK, AI)
blood_group VARCHAR(5)
units INT
urgency ENUM
status ENUM
■■ recipient_id → users.id
■■ hospital_id → hospitals.id

community_posts
■ id INT (PK, AI)
title VARCHAR(200)
content TEXT
blood_group_tag VARCHAR(5)
■■ donor_id → users.id

■ = Primary Key (PK) / **■■ = Foreign Key (FK → references)**
Arrows = 1:N Relationships (FK references)
ENUM = predefined value set · AI = Auto-Increment

14. Input & Output Design

14.1 Input Design

All user inputs in LifeFlow are collected through validated React 19 forms with real-time error feedback and smooth Framer Motion transitions:

- Registration Form: Name, email, password (Bcrypt-hashed on submit), blood group, contact number, role selection
- Donation Camp Booking: Camp selection from React-Leaflet map, preferred time slot, health declaration checkbox
- Emergency Request: Blood group required, quantity (units), hospital name, urgency level — submitted via Axios POST
- Blood Inventory Update (Hospital Admin): Blood group dropdown, unit count, expiry date — written to MySQL via Sequelize
- Community Story Post: Title, narrative text, blood group tag, optional image upload
- Search / Filter: City, blood group, date range — triggers Axios GET with query params to Express v5 API

14.2 Output Design

Outputs are delivered through responsive React 19 UI components with Tailwind CSS styling, tailored to each user role:

Section 14: Input & Output Design

14.1 Input Design

Registration Form
- ▶ Full name, Email
- ▶ Password (Bcrypt-hashed)
- ▶ Blood group, Contact
- ▶ Role selection

Camp Booking
- ▶ React-Leaflet map selection
- ▶ Preferred time slot
- ▶ Health declaration checkbox

Emergency Request
- ▶ Blood group required
- ▶ Quantity (units)
- ▶ Hospital name
- ▶ Urgency level (Axios POST)

Blood Inventory Update
- ▶ Blood group dropdown
- ▶ Unit count + Expiry date
- ▶ Written to MySQL/Sequelize

Community Story Post
- ▶ Title + narrative text
- ▶ Blood group tag
- ▶ Optional image upload

Search / Filter
- ▶ City, blood group
- ▶ Date range filter
- ▶ Axios GET → Express v5 API

14.2 Output Design

Donor Dashboard
- ▶ Donation history timeline
- ▶ XP & badge tier display
- ▶ Percentile ranking
- ▶ Lives saved counter

Blood Inventory Panel
- ▶ Live blood-group stock per hospital
- ▶ Critical shortage alerts (red)
- ▶ Supply-demand trend charts

Emergency Response
- ▶ Matched donor list with contact
- ▶ Real-time push notifications
- ▶ Request status tracker

Interactive Camp Map
- ▶ React-Leaflet markers
- ▶ Available slots counter
- ▶ Camp details popup

Leaderboard & Badges
- ▶ Top donors ranked by XP
- ▶ Badge tier: Starter→Legend
- ▶ Zustand real-time state

Admin Reports
- ▶ User & hospital management
- ▶ Analytics dashboard
- ▶ Exportable MySQL reports

15. Testing

LifeFlow underwent multiple phases of software testing to ensure reliability, security, and usability across all modules and user roles.

• Unit Testing

Individual components and API endpoints were tested in isolation using Vitest (React 19) and Mocha/Chai (Node.js). All 47 tested units passed with 94% code coverage.

• Integration Testing

Module interactions — especially the JWT + Bcrypt auth flow, MySQL queries via Sequelize, and real-time inventory sync — were tested end-to-end using Postman collections.

• System Testing

Complete user journeys (Registration → Camp Booking → Donation → Badge Award) were executed on staging environments mimicking production conditions.

• User Acceptance Testing (UAT)

Ten volunteer users performed predefined tasks. Feedback was used to refine UI/UX, particularly the Framer Motion animations, camp map filtering, and emergency request form.

• Security Testing

JWT token expiry, Bcrypt hash validation, role-based access control enforcement, SQL injection prevention, and CORS policy validation were all verified.

• Compatibility Testing

The application was tested on Chrome, Firefox, Safari, and Edge browsers, and on Android and iOS mobile browsers to ensure responsive Tailwind CSS rendering.

• Performance Testing

API response times were benchmarked under simulated load (100 concurrent users). Average response time remained under 320 ms. Vite build optimisation kept bundle size under 500 KB.

16. Future Enhancement

The following enhancements are planned for subsequent development phases of LifeFlow:

- Mobile Application: Native Android and iOS apps using React Native for wider donor reach and offline access
- Advanced AI Matching: ML model predicting optimal donor-recipient matches based on historical data and medical profiles
- Blockchain Blood Trail: Immutable donation history ledger using Ethereum smart contracts for transparency and trust
- Telemedicine Integration: Partnership with health platforms for pre-donation health screenings via telehealth APIs
- Multi-Language Support: Localisation for Hindi, Gujarati, and other regional languages to expand rural outreach
- Government API Integration: Real-time connection with National Blood Transfusion Council (NBTC) databases
- Predictive Analytics: AI-driven forecasting of blood demand spikes during festivals, accidents, or natural disasters
- Donor Health Wallet: Personal health card tracking haemoglobin levels, donation intervals, and health advisories
- WhatsApp / SMS Alerts: Automated emergency notifications via Twilio API for donors without smartphones

17. Conclusion

LifeFlow represents a significant step forward in digital health infrastructure for blood donation management in India. By combining real-time data, gamification, and community engagement into a unified React 19 + Node.js + MySQL platform, the project addresses a genuine humanitarian need while demonstrating advanced full-stack software engineering capabilities.

The platform successfully bridges the gap between willing donors and urgent recipients, reduces blood waste through accurate inventory tracking, and fosters a culture of voluntary regular donation through its innovative reward system. LifeFlow is designed not merely as an academic exercise but as a deployable, scalable solution ready for real-world adoption.

The project has strengthened the development team's expertise in React 19, Express v5 REST API design, relational database modelling with MySQL and Sequelize/Prisma, JWT + Bcrypt security, and user-centred design — skills directly transferable to professional software development roles.

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